COURSE TITLE: COMPUTER GRAPHICS AND VISUALISATION

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Introduction to Computer Graphics and Visualization

- **What is Computer Graphics?**

The term "**computer graphics**" refers to anything involved in the creation or manipulation of images on computer, including animated images. It is a very broad field, and one in which changes and advances seem to come at a dizzying pace.

In 1950 the first computer driven display, attached to MIT's computer, was used to generate simple pictures. This display used a Cathode-Ray tube (CRT). Interactive computer graphics made progress and the term computer graphics was first used in 1960.

It involves two ways communication between computer and user (Interactive Graphics Display). The computer receives signals from the input device, and modify the displayed picture appropriately. The main reason for the effectiveness of interactive computer graphics in many applications is the speed with which the user of the computer can assimilate the display information.

- **How it works?**

The modern graphics display is extremely simple in construction. It consists of three components:

- A digital memory, or frame buffer, in which the displayed image is stored as a matrix of intensity values. Frame buffer is the memory area that is used for storing picture definition as its stores the intensity values of the screen points.

- A monitor
- A display controller, which is a simple interface that passes the contents of the frame buffer to the monitor.

Inside the frame buffer the image is stored as a pattern of binary digital numbers, which represent a rectangular array of picture elements, or **pixel.**

The pixel is the smallest addressable screen element. In the simplest case where we wish to store only black and white images, we can represent black pixels by 0's in the frame buffer and white pixels by 1's. The display controller simply reads each successive byte of data from the frame buffer and convert each 0 and 1 to the corresponding video signal. This signal is then fed to the monitor. If we wish to change the displayed picture all we need to do is to change or modify the frame buffer contents to represent the new pattern of pixels.

▪ **Graphics Devices**

Typical examples are plotters, pen plotters, graphics tablet, light pen, cathode ray tube (CRT), raster scan, random scan, emissive display, non- emissive display, plasma panel display and LED&LCD.

✓ **Plotter**

The plotter is a computer printer for printing vector graphics. In the past, plotters were used in applications such as computer-aided design, though they have generally been replaced with wide-format conventional printers. A plotter gives a hard copy of the output. It draws pictures on a paper using a pen. Plotters are used to print designs of ships and machines, plans for buildings and so on.

Figure 1: Old Generation Plotter

Figure 2: Top View of a Plotter

- ✓ **Pen Plotters**

Pen plotters print by moving a pen or other instrument across the surface of a piece of paper. This means that plotters are vector graphics devices, rather than raster graphics as with other printers. Pen plotters can draw complex line art, including text, but do so slowly because of the mechanical movement of the pens. They are often incapable of efficiently creating a solid region of color, but can hatch an area by drawing a number of close, regular lines.

Plotters offered the fastest way to efficiently produce very large drawings or color high-resolution vector-based artwork when computer memory was very expensive and processor power was very limited, and other types of printers had limited graphic output capabilities.

Pen plotters have essentially become obsolete, and have been replaced by large-format inkjet printers and LED toner-based printers. Such devices may still understand vector languages originally designed for plotter use, because in many uses, they offer a more efficient alternative to raster data.

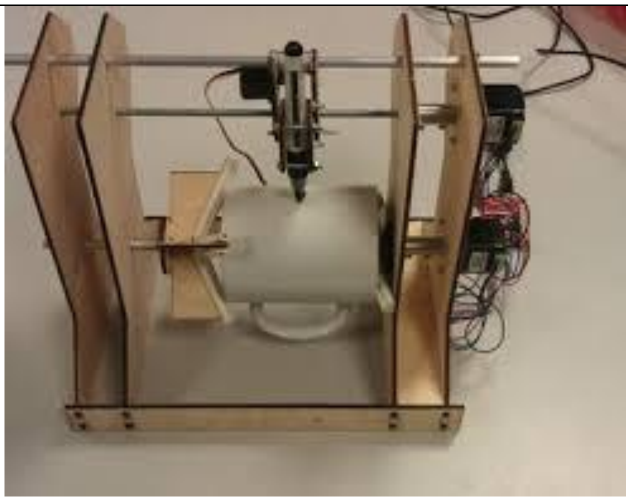
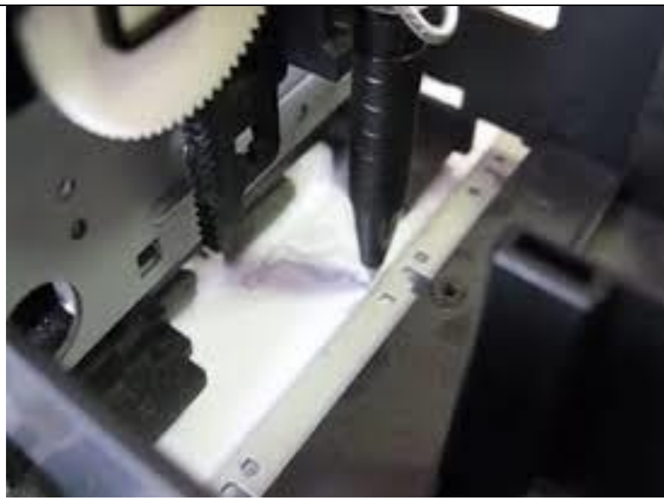



Figure 3: Pen Plotter

✓ Graphics Tablet

A graphic tablet also called **Digitizer**, **Drawing Tablet**, **Digital Drawing Tablet**, **Pen Tablet, Or Digital Art Board** is a computer input device that enables a user to hand-draw images, animations and graphics, with a special pen-like stylus, similar to the way a person draws images with a pencil and paper. These tablets may also be used to capture data or handwritten signatures. It can also be used to trace an image from a piece of paper which is taped or otherwise secured to the tablet surface. Capturing data in this way, by tracing or entering the corners of linear poly-lines or shapes, is called digitizing.

The device consists of a flat surface upon which the user may "draw" or trace an image using the attached stylus, a pen-like drawing apparatus. The image is displayed on the computer monitor, though some graphic tablets now also incorporate an LCD screen for a more realistic or natural experience and usability.

Figure 3: Graphics Tablet

Some tablets are intended as a replacement for the computer mouse as the primary pointing and navigation device for desktop computers. Below is a list of professions and people who are more likely to use a graphics tablet.

- Architects and Engineers
- Artists
- Cartoonist
- Fashion designers
- Graphic designers
- Illustrators
- Photographers
- Teachers

✓ Light Pen

A **light pen** is a computer input device in the form of a light-sensitive wand used in conjunction with a computer's CRT display. It allows the user to point to displayed objects or draw on the screen in a similar way to a touch screen but with greater positional accuracy. A light pen can work with any CRT-based display and other display technologies. A light pen detects a change of brightness of nearby screen pixels when scanned by cathode ray tube electron beam and communicates the timing of this event to the computer. Since a CRT scans the entire screen one pixel at a time, the computer can keep track of the expected time of scanning various locations on screen by the beam and infer the pen's position from the latest timestamp.

It is also worthy to mention that the Light Pen falls under the category of Raster Graphics.

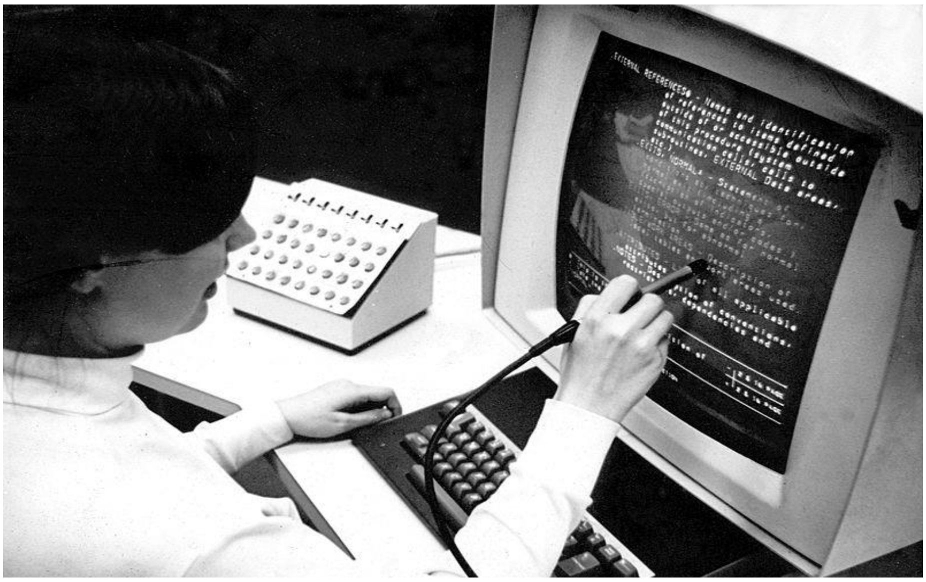


Figure 4: Light Pen

Cathode Ray Tube (CRT)

This is the first and the most common graphics device

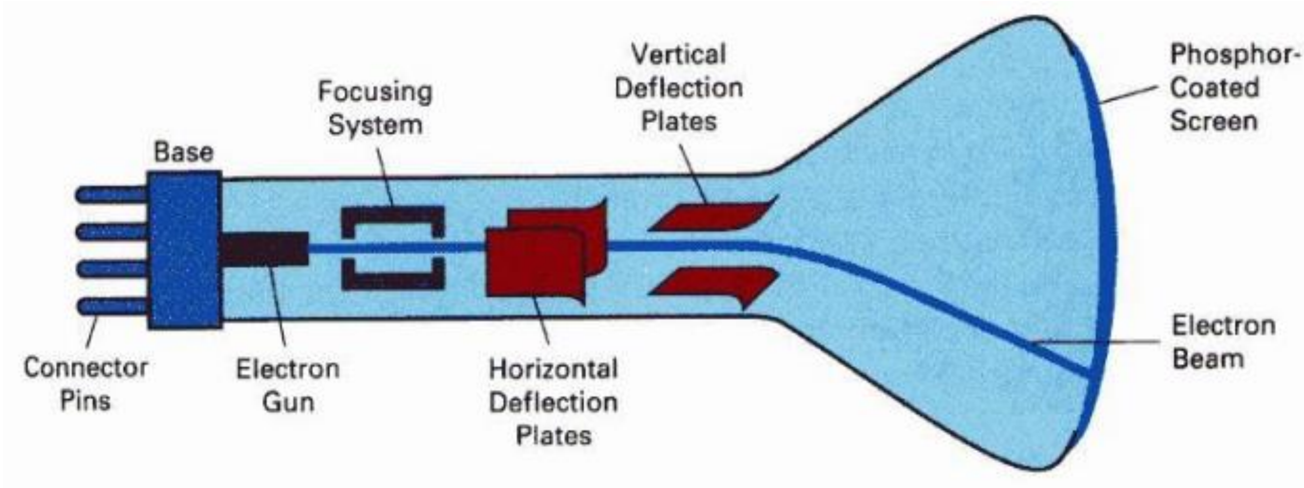
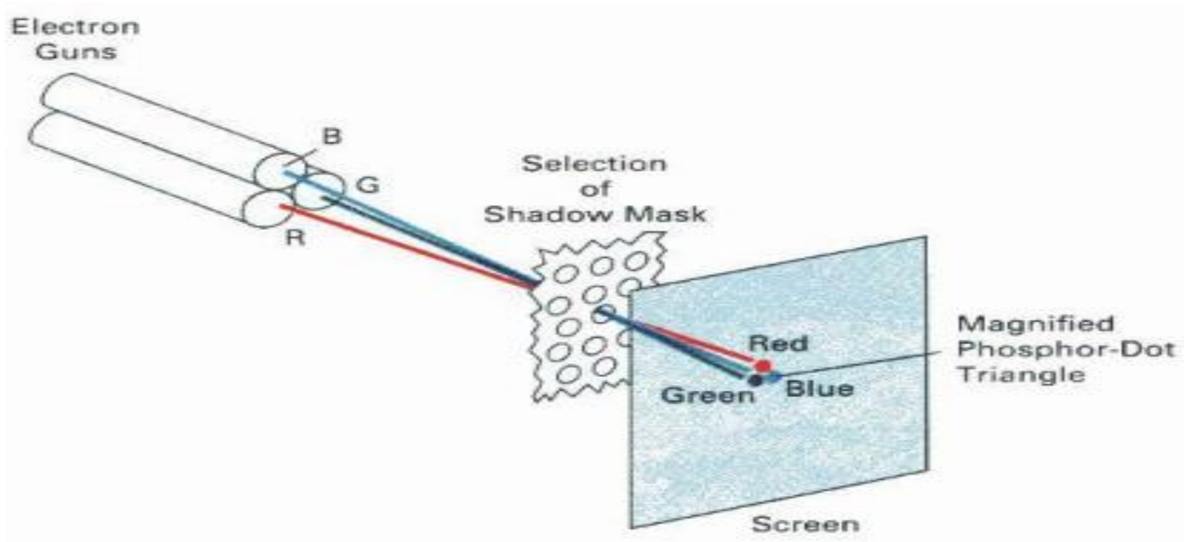


Figure 5: Cathode Ray Tube (CRT)

The Electron Gun emits a beam of electron which passes through the focusing and deflection system and hits the phosphor coated screen. The number of points displayed by the CRT is called Resolution. Different phosphors emit small light spots of different colors which combine to form range of colors.

Color CRT Display: - This is achieved using a methodology refer to as **SHADOW MASK MET**



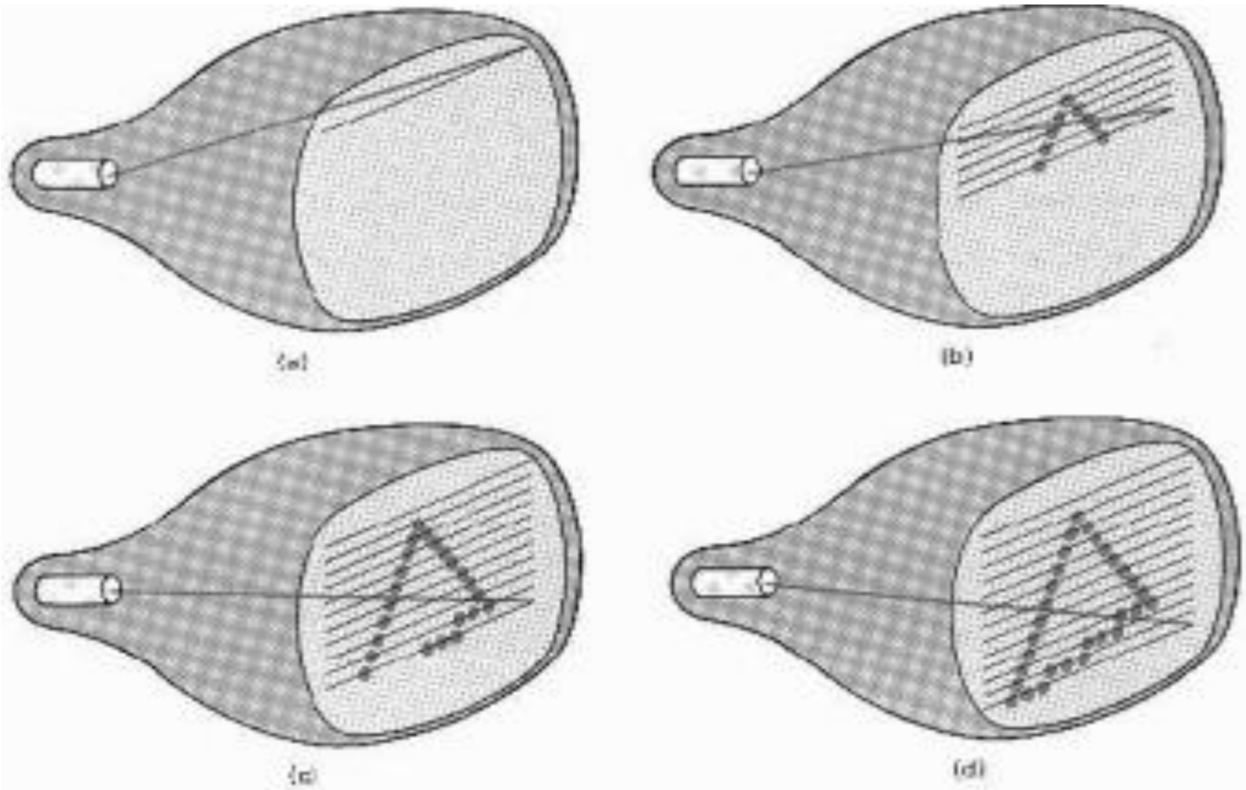
Here the Electron Gun emits three beams representing Red, Green and Blue (aka RGB) which passes through the Shadow Mask and falls on the phosphor to create the image. The light emitted by phosphor fade very quickly and thereby needs to be redrawn. Two (2) mechanisms were adopted to resolve this fading problem as follow:

A. Raster Scan or Raster Graphics
B. Random Scan or Vector Graphics

✓ **Raster Scan**

In Raster Scanning, the electron beam is swept across the screen one row at a time from top to bottom row. As it moves across each row, the beam intensity is turned ON and OFF to create a pattern for the object. This rowise scanning process is called **REFRESHING**. Each complete scanning of a screen is normally called a **FRAME**.

The **REFRESHING RATE** is called **FRAME RATE** and it is normally 60 to 80 frames per second, otherwise called 60Hz to 80Hz.

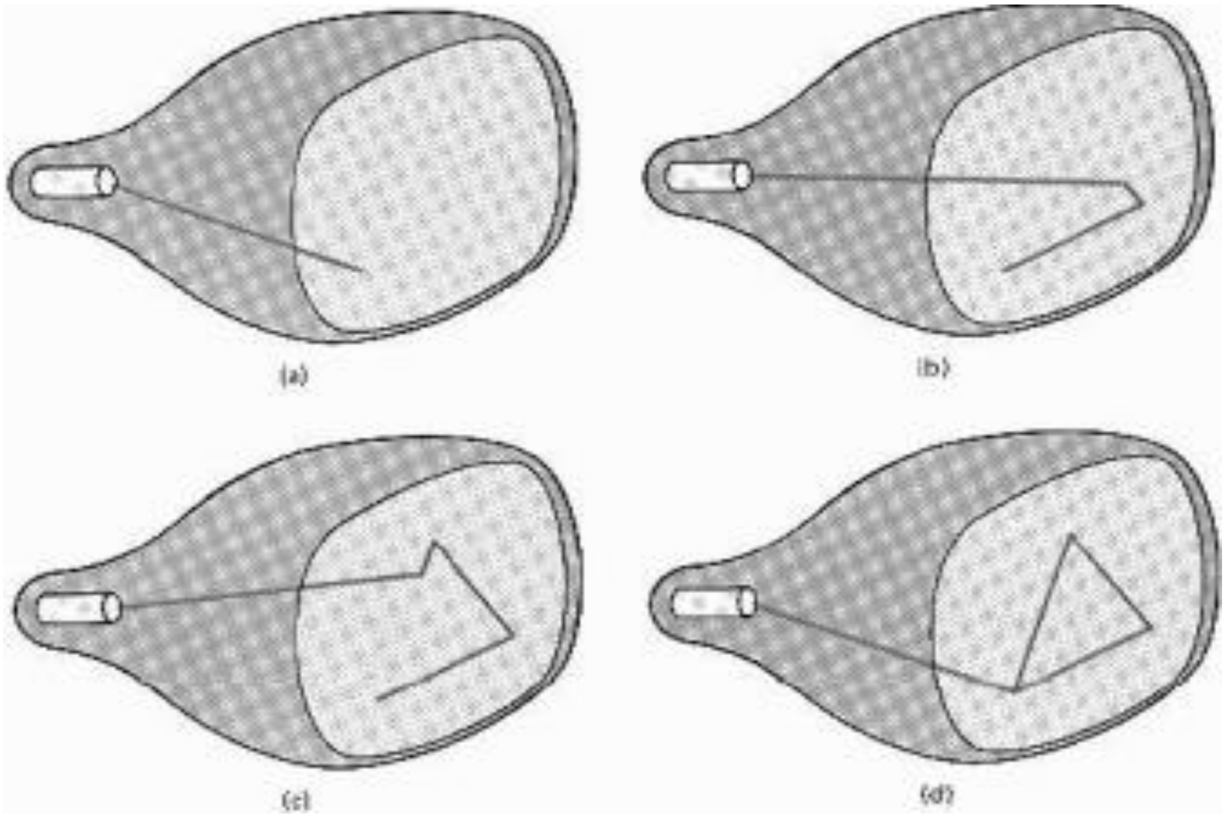


FRAME BUFFER: - Is the memory area that is used for storing picture definition as its stores the intensity values of the screen points.

PIXEL: - Refers to Picture Element which represents screen point.

- ✓ **Random Scan**

In Random Scan otherwise called Vector Display, the CRT’s electron beam is directed only to the parts of the screen where a picture is to be drawn. The Picture definition is stored as a set of line-drawing command in a refresh display file called Refresh Buffer in memory.



Raster Graphics or **Bitmap** Image is a dot matrix data structure, representing a generally rectangular grid of pixels, or points of color, viewable via a monitor, paper, or other display medium. Raster images are stored in image files with varying formats.

A bitmap is a single-bit raster that correspond bit-for-bit with an image displayed on a screen, generally in the same format used for storage in the display's video memory, or maybe as a device-independent bitmap. A raster is technically characterized by the width and height of the image in pixels and by the number of bits per pixel (or color depth, which determines the number of colors it can represent).

- ✓ **Emissive Display**

The emissive displays are devices that convert electrical energy into light. Examples are Plasma Panel, thin film electroluminescent display and LED (Light Emitting Diodes).

- ✓ **Non-Emissive Display**

The Non-Emissive displays use optical effects to convert sunlight or light from some other source into graphics patterns. Examples are LCD (Liquid Crystal Device).

The Flat-Panel display refers to a class of video devices that have reduced volume, weight and power requirement compare to CRT. Examples include Small T.V. monitor, calculator, pocket video games, laptop computers, an advertisement board in elevator.

- ✓ **Plasma Panel Display**

Plasma-Panels are also called as Gas-Discharge Display. It consists of an array of small lights. Lights are fluorescent in nature. The gas will slow when there is a significant voltage difference between horizontal and vertical wires. The voltage level is kept between 90 volts to 120 volts. Plasma level does not require refreshing. Erasing is done by reducing the voltage to 90 volts.

Each cell of plasma has two states, so cell is said to be stable. Displayable point in plasma panel is made by the crossing of the horizontal and vertical grid. The resolution of the plasma panel can be up to 512 * 512 pixels.

✓ **LED & LCD**

In an **LED,** a matrix of diodes is organized to form the pixel positions in the display and picture definition is stored in a refresh buffer. Data is read from the refresh buffer and converted to voltage levels that are applied to the diodes to produce the light pattern in the display.

Liquid Crystal Displays are the devices that produce a picture by passing polarized light from the surroundings or from an internal light source through a liquid-crystal material that transmits the light.

LCD uses the liquid-crystal material between two glass plates; each plate is the right angle to each other between plates liquid is filled. One glass plate consists of rows of conductors arranged in vertical direction. Another glass plate is consisting of a row of conductors arranged in horizontal direction. The pixel position is determined by the intersection of the vertical & horizontal conductor. This position is an active part of the screen.

Liquid crystal display is temperature dependent. It is between zero to seventy degree Celsius. It is flat and requires very little power to operate.

▪ **Input & Output Devices**

- **The Input Devices** are the hardware that is used to transfer input to the computer. The data can be in the form of text, graphics, sound, and text. Examples Include;
  - Keyboard, Mouse, Trackball, Spaceball, Joystick, Light Pen, Digitizer, Touch Panels, Voice Recognition (microphone) and Image Scanner.

- **Output device** display data from the memory of the computer. Output can be text, numeric data, line, polygon, and other objects. Examples include
  - Printers, Plotters and Speakers.

- **Visualization**

- This simply means data/information transformation into pictures. This does not necessarily imply the use of computers.
- Classical visualization used hand-drawn figures and illustrations (2D means for visualization) Modern visualization is primarily 3D (digital images for 3D visualization)
- In both cases, the ultimate goal is to understand important insights about the data through ***visual means*** We really don't care ***how we get the picture in visualization –what picture we get is most important***
- Typical of a visualization application is the field of Computer Graphics The invention of computer graphics may be the most important development in visualization, Thus **Visual Computing**
- Visual Computing is the field of acquiring, analysing, processing, and synthesizing of visual data by means of computers.
- Visual data–includes data from other sources, such as financial, marketing, business. Sometimes involves statistical analysis and other analysis techniques not employed in scientific visualization

- Visualization today has ever-expanding applications in science, education, engineering (e.g., product visualization), interactive multimedia, medicine etc.

- **Why do we Visualize?**
- Important features and meaningful knowledge of the data to assist in the decision-making process.

- Reduce time and save money.
- Process of exploring, transforming and viewing data as images.
- Visualization may employ graphics to generate images.
- Visualization may employ image processing to study images and the process involves;
    - Data acquisition
    - Data transformation
    - Data mapping (e.g., to shapes & color)
    - Display (via computer graphics).

- **Applications of Computer Graphics & Visualisation**

✓ **Display of Information**

Graphics has always been associated with the display of information. Examples of the use of orthographic projections to display floor plans of buildings can be found on 4000-year-old Babylonian stone tablets.

In fields such as architecture and mechanical design, hand drafting is being replaced by computer-based drafting systems using plotters and workstations. Medical imaging uses computer graphics in a number of exciting ways.

Mechanical methods for creating perspective drawings were developed during the Renaissance. Countless engineering students have become familiar with interpreting data plotted on log paper. More recently, software packages that allow interactive design of charts incorporating color, multiple data sets, and alternate plotting methods have become the norm.

✓ **Design**

Professions such as engineering and architecture are concerned with design. Although their applications vary, most designers face similar difficulties and use similar methodologies.

One of the principal characteristics of most design problems is the lack of a unique solution. Hence, the designer will examine a potential design and then will modify it, possibly many times, in an attempt to achieve a better solution. Computer graphics has become an indispensable element in this iterative process.

Computer Graphics may be use to design an electronic circuit. The designer is seated at a graphics workstation with a graphical input device, such as a mouse, with which she can indicate locations on the display.

The initial display screen might consist of the various elements that can be used in the circuit and an empty area in which the circuit will be "constructed." The designer will then use the input device to select and move the desired elements into the design and to make connections between elements.

To form this initial design, the system makes sophisticated use of computer graphics. Circuit elements are drawn, and perhaps are moved about the screen. A graphical input device is used to indicate choices and positions.

A number of aids may be used to help the designer position the elements accurately and to do automatically such tasks as routing of wires.

- ✓ **Simulation**

Computer-generated images are also the heart of flight simulators, which have become the standard method for training pilots. The savings in dollars and lives realized from use of these simulators has been enormous. The computer-generated images we see on television and in movies have advanced to the point that they are almost indistinguishable from real-world images.

- ✓ **User Interfaces**

The interface between the human and the computer has been radically altered by the use of computer graphics. Consider the electronic office. The figures in this book were produced through just such an interface. A secretary sits at a workstation, rather than at a desk equipped with a typewriter. This user has a pointing device, such as a mouse, that allows him to communicate with the workstation. The display consists of a number of icons that represent the various operations the secretary can perform. For example, there might be an icon of a mailbox

- ✓ **Graphics API**

An application program interface (API) is a standard collection of functions to perform a set of related operations, and a graphics API is a set of functions that perform basic operations such as drawing images and 3D surfaces into windows on the screen.

Every graphics program needs to be able to use two related APIs: a graphics API for visual output and a user-interface API to get input from the user. There are currently two dominant paradigms for graphics and user-interface APIs.

- The first is the integrated approach, exemplified by Java, where the graphics and user interface toolkits are integrated and portable packages that are fully standardized and supported as part of the language.
- The second is represented by Direct3D and OpenGL, where the drawing commands are part of a software library tied to a language such as C++, and the user-interface software is an independent entity that might vary from system to system.

- ✓ **Popular Applications Used for Computer Graphics and Visualisation**

- Adobe photo shop
- Corel Draw
- Blender

Transformation

Basic transformation includes:

- Translation
- Rotation
- Scaling
- Reflection
- Shearing

We will look two each of these transformations with examples

1. Translation: This is the shift or change in location of an object form one point to another. i.e from point (x, y) to point ($x^{|}$, $y^{|}$). To do this shift we need a shifting factor or a distance vector known as the shift vector such that.

   $x^{|} = x + Tx$ where Tx and Ty are the shift vector

   $y^{|} = y + Ty$

example: Given the 2D object

a(2, 3), b (10, 4), c(6, 6) translate with Tx = 5 and Ty = 6

so basically going by formular, we go for each point

$a^{|} = a + \vec{T}$

$a^{|} = \binom{2}{3} + \binom{5}{6} = \binom{7}{9}$

$b^{|} = \binom{10}{4} + \binom{5}{6} = \binom{15}{10}$

$c^{|} = \binom{6}{6} + \binom{5}{6} = \binom{11}{12}$

$a^{|}$ (7,9), $b^{|}$ (15, 10), $c^{|}$ (11, 12)

the same also applies to 3D-objects such that the shift factor:

$$Tv = \begin{cases} x^{|} = x + 4x \\ y^{|} = y + Ty \\ z^{|} = z + Tz \end{cases}$$

along x, y and z axis. In this instance we have to use a homogeneous form of matrix such that

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & Tx \\ 0 & 1 & 0 & Ty \\ 0 & 0 & 1 & Tz \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ homogeneous factor}$$

Example a point has a coordinate (x, y, z) direction 8, 9, 10 with translation vector in x direction $T_x = 3$ $T_y = 3$ and $T_z = 2$. Determine the direction

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 8 \\ 9 \\ 10 \\ 11 \end{bmatrix} = \begin{bmatrix} 8+0+0\ \ 0 & 3 \\ 0\ +\ 9\ +0\ + & 3 \\ 0+0\ +10+ & 2 \\ 0+0+\ 0\ + & 1 \end{bmatrix} = \begin{bmatrix} 11 \\ 12 \\ 12 \\ 1 \end{bmatrix}$$

So we recalibrate the position from point 8, 9, 10 to point 11, 12, 12 for x, y and z respectively.

2. Scaling

This is used to change the size of the object. The scaling factors are

$Sx \longrightarrow$ x direction

$Sy \longrightarrow$ y direction such that

Scaling is also defined as the process of expanding or compressing the dimension of an object.

Scale (Sx, Sy) = $\begin{bmatrix} Sx & 0 \\ 0 & Sy \end{bmatrix}$

Such that $\begin{bmatrix} Sx & X \\ Sy & Y \end{bmatrix} = \begin{bmatrix} Sx & 0 \\ 0 & Sy \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \longrightarrow$ $\boxed{P^I = S.P}$

If scaling factor = 1 object size will not change

> 1 enlarge

< 1 compresses or reduces

Example: scale the object with coordinates A (2,5), B (3, 10), C (10, 3) by 2 units in x-direction and 3 units in y– direction.

$\longrightarrow$ $S_x = 2$, $S_y = 3$ Given our formation ($P^I = S.P$)

Q. $\begin{bmatrix} 2 & 5 \\ 3 & 10 \\ 10 & 3 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

$\begin{bmatrix} 4 & 15 \\ 6 & 30 \\ 20 & 9 \end{bmatrix} \longrightarrow$ New Scale $A^I (4, 15)$

$B^I (6, 30)$

$C^I (20, 9)$

For scaling in 3D, the scaling factors are Sx, Sy and Sz along x, y and z axis respectively. Similar 3D translation we use a homogeneous factor within the matrix such that

$$P^I = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} Sx & 0 & 0 & 3 \\ 0 & Sy & 0 & 3 \\ 0 & 0 & Sz & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

3. ROTATION

This transformation involves the process of changing the angle of the object. This can be done both in clockwise direction and anti-clockwise direction.

Let coordinate P in the Cartesian coordinate of an object be

$$x = rcos\Theta \; and \; y = rsin\Theta$$

$$x = rcos\Theta$$

$$y = rsin\Theta \qquad x' = r[cos\Theta \, cos\Theta - sin\Theta sin\Theta]$$

$$= r \, cos\theta \, cos\Theta - rsin\Theta \, sin\Theta$$

Similarly, $X' = xcos\Theta - \text{y} \, sin\Theta$

$$Y' = ycos\Theta - \text{x}sin\Theta$$

So $\begin{bmatrix} x' \\ y \end{bmatrix} = \begin{bmatrix} [cos\Theta - & sin\Theta \\ sin\Theta & cos\Theta \end{bmatrix} \begin{bmatrix} x' \\ y \end{bmatrix}$ P’ = R. P where R is the factor rotational

Example: given the triangle a (2,2) b (9, 2), c (6, 6)

Rotate the angle by 90^0

$$\theta = 90^0$$

$$\begin{bmatrix}[cos\Theta - & sin\Theta \\ sin\Theta & cos\Theta\end{bmatrix} = \begin{bmatrix}0 & -1 \\ 1 & 0\end{bmatrix}$$

So, we solve for each part since R = $\begin{bmatrix}0 & -1 \\ 1 & 0\end{bmatrix}$

For a point a $\begin{bmatrix}x' \\ y'\end{bmatrix} = \begin{bmatrix}0 & -1 \\ 1 & 0\end{bmatrix}\begin{bmatrix}2 \\ 2\end{bmatrix} = \begin{bmatrix}-2 \\ 2\end{bmatrix}$

b $\begin{bmatrix}x' \\ y'\end{bmatrix} = \begin{bmatrix}0 & -1 \\ 1 & 0\end{bmatrix}\begin{bmatrix}9 \\ 2\end{bmatrix} = \begin{bmatrix}-2 \\ 2\end{bmatrix}$

c $\begin{bmatrix}x' \\ y'\end{bmatrix} = \begin{bmatrix}0 & -1 \\ 1 & 0\end{bmatrix}\begin{bmatrix}6 \\ 6\end{bmatrix} = \begin{bmatrix}-6 \\ 6\end{bmatrix}$

a (-2, 2) b (-2, 9) c (-6, 6)

3D ROTATION

At the initial co-ordinate of an object on 3D plane can be P (x, y, z) also initial angle Θ such that rotational angle be Θ

New co-ordinate after rotation

P' (x', y', z')

Note that rotation is done with respect to the direction of the object is inclined at i.e x-axis, y-axis and z-axis.

For x-axis:

x' = x

y' = $ycos\Theta - \mathrm{z}sin\Theta$

z' = $ysin\Theta - \mathrm{zcos}\Theta$

For y-axis:

x'= $zsin\Theta - \mathrm{z}cos\Theta$

y'= y

z = $zcos\Theta - \mathrm{z}sin\Theta$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & Cos\Theta & sin\Theta & 0 \\ 0 & sin\Theta & cos\Theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \qquad \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} cos\Theta & 0 & sin\,\Theta & 0 \\ 0 & 1 & 0 & 0 \\ 0 & sin\Theta & cos\Theta & 0 \\ & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

For z-axis:

x’= $ycos\Theta - ysin\Theta$

y’ = $xsin\Theta - \mathrm{ycos}\Theta$

z’ = z

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\Theta & sin\Theta & 0 & 0 \\ sin\Theta & Cos\Theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Example:

Given the points p (1, 4, 6) rotate by 90^0

x-axis:

x' = x = x’ = 1

y’ = $ycos\Theta - \mathrm{z}sin\Theta$

4 (0) – 6 (1)

= -6

z’ = $ysin\Theta + \mathrm{zcos}\Theta$

4 (1) + 6 (0)

= 4

along x-axis

For y-axis:

x’= $zsin\Theta + cos\Theta$

6(1) +1 (0)

6

y’ = y = 4

z’ = $zcos\Theta - \mathrm{x}sin\Theta$

6 (0) – 1 (1)

= -1

along y-axis

p’ = (1, -6, 4) p(6, (4), (-1)

along z-axis:

x’= $ycos\Theta - ysin\Theta$

= 0 – 4

= - 4

y’ = $xsin\Theta + y\cos\Theta$

1 (1) + 0

=1

z’ = z = 6

P (-4, 1, 6) along z-axis

4.SHEARING

Shear is something that pushes things sideways producing something like a deck of cards across which you push your hands. The bottom cards stays put and cards move more closer to the top of the deck.

Sliding layer in our direction to change the shape of the object.

X-SHEARING

here the y-coordinates remain the same since our slide is in x-direction P(x,y) new point p’(x’,y’)

x’ = x + shx.y

y’ = y

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} y & shx \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$ or x-changes

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ shx & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \longrightarrow$$ homogeneous coordinate

Y-SHEARING

Here the x-coordinates remain the same since the sliding is in y-direction

$$x' = x$$

$$y' = y + shy.x$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ shy & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ shx & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

The matrix $\begin{bmatrix} 1 & a \\ b & 1 \end{bmatrix}$ defines a transformation called simultaneous given the square A(0,0) B(1,0) C(1,1) D(0,1) determine the (i) sh.x (ii) sh.y (iii) simultaneous sh.

(i) $\begin{bmatrix} x' \\ y \end{bmatrix} = \begin{bmatrix} 1 & shx \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$ (ii) $\begin{bmatrix} x, \\ y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ shy & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$ (iii) $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

A = $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ A = $\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ A = $\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

B = $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ B = $\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ B = $\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

C = $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ C = $\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ C = $\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

D = $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ D = $\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ D = $\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

5.REFLECTION

This transformation tries to capture the mirror image of an object. 2D-reflection transformation is done with respect to x-axis and y-axis

So for instance Re(x) of point p(x,y)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$p' \rightarrow Re(x).p$$

Re(y)

$$\begin{bmatrix} x' \\ y \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$p' \rightarrow Re(y).p$$

Example: Given the coordinate points of a given triangle A (3, 5) B(7,2) C(4,8) find the reflected position on x-axis

$p' = Re(x).p$ so we solve for each point

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

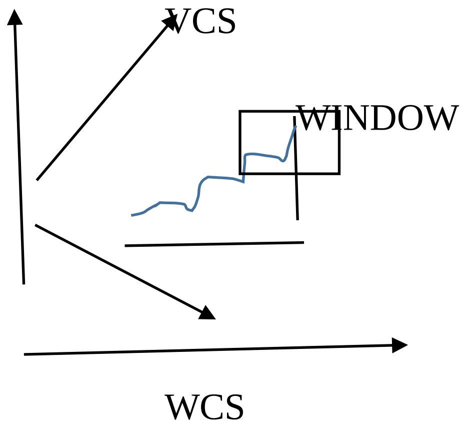
$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 7 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 4 \\ -8 \end{bmatrix}$$

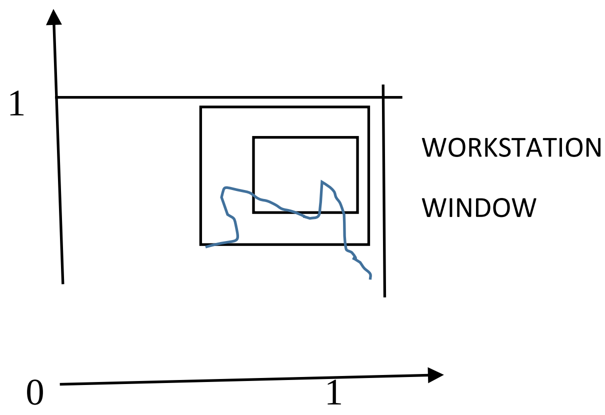
- **VIEWING AND CLIPPING**

World coordinate systems: Objects are placed into a scene by modelling transformations to master coordinate systems. This coordinate systems are termed world coordinate systems (wcs); basically, used to help window select the portion of the scene for which the image is generated.

- VCS is viewing coordinate systems are coordinates introduced to simulate the effect of viewing and / or tilting the camera.



- NDCS- Normalized coordinate system (NDCS) is a system which, is a unit (1 x 1) square whose lower left corner is at the origin of the coordinate system.



Clipping: This implies cutting of the unrequired part. The clip operation eliminates objects or portion of object not visible through the window to ensure proper construction of corresponding image.

- **POINT CLIPPING**

This is to determine of a specific point lies within the window or not such that

$X_{min} \le x \le X_{max}$ $Y_{min} \le y \le Y_{max}$

Where Xmin, Xmax, Ymin and Ymax define the clipping window A point (x, y) is considered inside the window when the inequalities all evaluate to true.

- **LINE CLIPPING**

Lines that do not intersect the window are either completely inside the window or outside. Lines that do intersect the clipping window is however divided by the intersection points into segments that are either inside or outside the window.

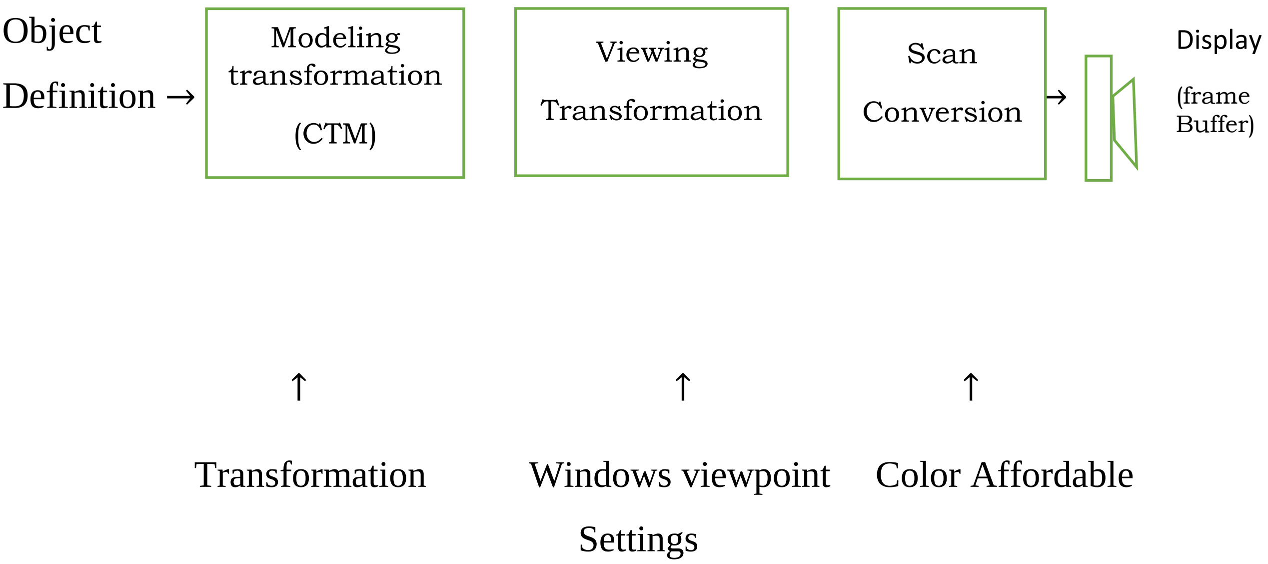
Those inside the window = visible

Outside the window = Not visible

Inside/outside = particularly visible (clipping candidates)

- **2D-GRAPHICS PIPELING**

The sequence of operations that is required when processing a primitive object to its masterized form. i.e starting with objects (primitive object) and ending by updating pixels in the image is known as graphics pipeline. Although 2D in the image is typically treated as a special case (Z=0).



Whenever a drawing subroutine is called to render a predefine object, the graphics system first applies the specified modeling transformation to the original data, then carries out viewpoint setting and finally perform a scan conversion to set the proper pixel in the frame buffer with the specified color attributes.

- **SCAN CONVERSION**

The conversion of primitive objects (points, circle lines etc) from its geometric definition into a set of pixels by a graphic system (Application program) that shows a representation of the primitive in the image space is referred to as scan conversion of Rasterization.

- **DIGITAL DIFFERENTIAL ANALYZER (DDA)**

This is a line drawing algorithm. It is a scan conversion method which uses the incremental approach. This implies calculations are done at each step using results from preceeding step.

Considering the straight line equation

$y = mx+b$

if u line has two points $A(x_1y_1)$ and $B(x_2y_2)$

slope of the line is m= y_2-y_1

x_2-x_1

$$m = \frac{\Delta y}{\Delta x}$$

So the DDA is based on the calculation of values of $\Delta y \; and \; \Delta x$

Given $m = \frac{\Delta y}{\Delta x}$ $\quad \Delta y = m\Delta x$

$$\Delta y = y_2 - y_1 \qquad \therefore y_2 = y_1 + m\Delta x$$

Similarly $x_2 = x_1 + \frac{\Delta x}{m}$

These formulars are used in DDA algorithm in this way

Note $m > 0$ +ve slope $\quad \Delta x \; and \; \Delta y$ values are increased

$m \leq 0$ –ve slope $\quad \Delta x \; and \; \Delta y$ values are decreased

Case 1: if $m > 0$	Case 2: If $m > 1$	Case 3: If $m =1$
$x_n = x_1 + 1$	$x_n = x_1 + \frac{1}{m}$	$x_n = x_1 + 1$
$y_n = y_1 + 1$	$y_n = x_1 + 1$	$y_n = y_1 + 1$

- **ADVANTAGES OF DIGITAL DIFFERENTIAL ANALYZER (DDA)**
    - Faster when compared to direct line equation.
    - Multiplication theorem not required
    - Does not allow plotting of same point twice
    - Allow us to detect when a point is positioned.
    - It is easy as each step only involves addition.

DISADVANTAGES OF DIGITAL DIFFERENTIAL ANALYZER (DDA)

- Floating point additional and rounding off leads to error.
- Rounding off and floating point operation consumes a lot of time.
- Software implementation better than hardware implementation.